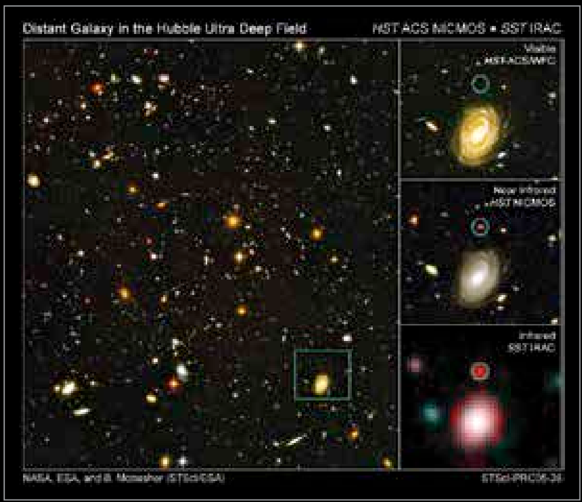

Images captured in Infrared can capture detail not visible if captured using light in the visible spectrum

to operate much farther away from the Earth than the Hubble operates, it also has a massive solar shield (the size of a tennis court!) that keeps the mirrors and the instruments at optimum functioning temperature. The combination of a distant orbit and an efficient sunshield gives the JWST a massive advantage of being able to sense light of higher wavelengths (~4 x HST), falling in the infra-red (IR) spectrum. This means that while signals falling in the visible range of wavelengths will lead to images equally sharp as that captured by the HST, the JWS, with its enhanced sensitivity and larger mirror will produce more detailed images when a signal falls in the IR spectrum.

Why is the JWST's IR sensing capability such a big deal?

Our Universe is expanding, stretching space time along the way, moving galaxies away from us. Light coming from these fast moving heavenly bodies is 'stretched' by the time it reaches our telescopes. Light coming from a galaxy that formed very early (and is deeper out in space) will hence be more stretched than light coming from a galaxy only recently formed. This 'stretching' effectively serves to increase the wavelength of light in a phenomenon called the Doppler shift. If we want to look at a galaxy that formed shortly after the birth of the universe our telescopes need to be able to capture light that is stretched to such an extent that after starting out as a UV ray, it bypasses the visible spectrum and falls into...you guessed it, the IR range! The JWST has the ability to sense light falling in the IR and near IR range, which means that it can look further back in time, farther out in space, and thanks to its giant mirror, with much greater clarity. The long wavelength of IR has certain benefits. It is not obscured by dust as much as smaller wavelengths of radiation. The birth of stars, planets and many planetary bodies are seeded by dust clouds filled with granular particles that block radiation falling in the visible spectrum. A telescope like the JWST that is optimized for picking up IR radiation however, can sense the IR radiation that these bodies glow with, giving scientists the ability to study the formation of planets and stars with such detail that simply isn't possible with any other space or Earth Bound telescopes.

What can and what does the JWST want to achieve?

In the previous passage we established that the JWST can collect data from bodies in the early universe. How far back are we talking about? The HST with its limited sensing capabilities can observe galaxies that are close to 500 million years old. The JWST can detect light coming from a galaxy that is almost 13.5 billion years old. These galaxies would have formed at a time when the universe was just about 1-2 per cent of its current age. That is a phenomenal advance! In fact the primary goal of the JWST is to study the early universe which is also the distant universe. Its ability to look through large dust dominated clouds means that it can study objects that were till date primarily obscured due to our limited ability to sense IR. These regions include molecular clouds which are usually where stars take birth, disk shaped regions that seed planet formation and even the core of galaxies. More importantly it can study these regions when they first formed, at the very infancy of our universe! The JWST can not only image these regions but also analyze the chemistry surrounding these interstellar systems by doing an advanced spectral analysis and breaking down an incoming signal into its constituent wavelengths. Using this tool the JWST plans to study various exoplanets and even ascertain possibilities of candidate planets harboring life. The primary equipment aboard the JWST include a near infrared spectrograph and a slitless spectrograph. Both of these instruments complement the infrared camera and the mid infrared instrument, the primary imaging technology that will be responsible for most of the dazzling images that the telescope captures. The wide range of wavelengths that can be captured by the JWST also allow it another luxury – the ability to study 'cool' objects. Cold objects emit less energetic and consequently higher wavelength radiation. Many interesting interstellar objects become cool objects in their lifetime. Examples include comets and even certain planets in galaxies scattered across the universe. Indirectly, the JWST can contribute to dark matter and dark energy research as well and will likely be

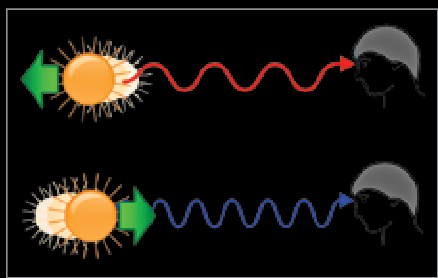

Image showing how light coming from objects receding from us (like galaxies) will have a larger wavelength shifted towards red end in the electromagnetic spectrum

used as valuable tool to pursue that line of research. Light coming from distant galaxies gets bent due to the effect of the mass of near galaxies (Einstein's theory of relativity dictates that space time is disturbed in the presence of mass). Thus, the image obtained of distant galaxies will be shifted. This effect, called gravitational lensing, allows scientists to measure the actual mass of the near galaxy and contrast it with the mass calculated by an approximate summing of luminous objects.